

MATH 3150 — Quiz 1 Study Guide

Methods, templates, worked examples, and practice problems

(Weeks 1–4: $\mathbb{N}, \mathbb{Q}, \mathbb{R}$, induction, sup/inf, completeness, absolute value, limits of sequences, monotone sequences)

Quiz 1: Tuesday, September 29

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1 How to use this guide

Each section follows the same pattern: **definitions you must be able to state word for word**, then **the method**, then **worked examples**. Section 10 has practice problems, and their solutions are in Section 11. Try each problem on paper before you look at the solution.

Quiz scope (from the syllabus).

- HW1: set identities, induction, rational/irrational arguments.
- HW2: density of $\mathbb{Q}$ and of the irrationals, sup/inf, the completeness axiom, $\sup(A + B)$.
- HW3: triangle and reverse triangle inequality, ε - N limits, $x_n \rightarrow \infty$, monotone sequences.
- Week 4 lectures: limit laws, basic limits, limits equal to $\pm\infty$, the Monotone Convergence Theorem.

The single most important skill: match the *shape* of the claim to the proof technique.

Claim looks like...	Reach for...
$S = T$ (sets)	Double inclusion, element chasing
“for all $n \geq n_0$ ”	Induction
“ x is irrational”, “no such ...”	Contradiction
“is it always true?” (suspect false)	One counterexample
$M = \sup S$	(i) M is an upper bound, (ii) $\forall \varepsilon > 0 \exists s \in S, s > M - \varepsilon$
$\sup A \leq \text{something}$	Show that “something” is an upper bound of A
$\lim x_n = L$	ε - N : scratch work, then a forward write-up
$\lim x_n = \infty$	$\forall M \exists N: x_n > M$ for $n \geq N$
x_n converges (limit unknown)	Monotone Convergence Theorem
$ a \leq c$	Show $-c \leq a \leq c$

Definitions cheat sheet (memorize word for word)

Upper bound	M with $s \leq M$ for all $s \in S$
$\sup S$	upper bound M such that no $M' < M$ is an upper bound $\iff$ upper bound and $\forall \varepsilon > 0 \exists s \in S : s > M - \varepsilon$
$\inf S$	lower bound m such that no $m' > m$ is a lower bound $\iff$ lower bound and $\forall \varepsilon > 0 \exists s \in S : s < m + \varepsilon$
Completeness	nonempty + bounded above $\Rightarrow$ sup exists in $\mathbb{R}$
Archimedean	$\forall x \in \mathbb{R} \exists n \in \mathbb{N} : n > x$; equivalently $\forall \varepsilon > 0 \exists n : \frac{1}{n} < \varepsilon$
Density	$x < y \Rightarrow \exists r \in \mathbb{Q}$ and $\exists z \notin \mathbb{Q}$ with $x < r, z < y$
$ x \leq a$	$\iff -a \leq x \leq a$ (for $a \geq 0$)
$x_n \rightarrow L$	$\forall \varepsilon > 0 \exists N \forall n \geq N : x_n - L < \varepsilon$
$x_n \not\rightarrow L$	$\exists \varepsilon > 0 \forall N \exists n \geq N : x_n - L \geq \varepsilon$
$x_n \rightarrow \infty$	$\forall M \exists N \forall n \geq N : x_n > M$
Bounded seq.	$\exists M > 0 \forall n : x_n \leq M$
Increasing	$x_n \leq x_{n+1}$ for all n (decreasing: $x_n \geq x_{n+1}$)
MCT	bounded + monotone $\Rightarrow$ convergent (to sup or inf of the terms)
Cauchy	$\forall \varepsilon > 0 \exists N \forall n, m \geq N : x_n - x_m < \varepsilon$

2 Logic and proof techniques

2.1 The basic proof moves

- **Direct:** assume the hypotheses and chain implications to the conclusion.
- **Contrapositive:** $P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$. Example: “ n^2 even $\Rightarrow n$ even” is easiest as “ n odd $\Rightarrow n^2$ odd”.
- **Contradiction:** assume $\neg(\text{claim})$ and derive something false.
- **Proving “ $a \leq b$ ” via ε :** if $a < b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$. (Otherwise take $\varepsilon = a - b > 0$.) Likewise, if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$. **This is how uniqueness of limits and many sup facts are proved.**
- **“For every” statements:** begin with “Let ... be arbitrary.” **“There exists” statements:** construct or exhibit the object.

2.2 Negating quantifiers

To negate a statement, swap $\forall \leftrightarrow \exists$ and negate the inner condition.

$$\neg(\forall \varepsilon > 0 \exists N \forall n \geq N : |x_n - L| < \varepsilon) \iff \exists \varepsilon > 0 \forall N \exists n \geq N : |x_n - L| \geq \varepsilon.$$

So “ $x_n \not\rightarrow L$ ” means: *some* ε is a bad tolerance, and the sequence keeps returning to distance $\geq \varepsilon$ from L no matter how far out you go. You use this to prove divergence (Example 5.8).

2.3 Double inclusion (HW1 P1)

Method 2.1. To prove $S = T$, prove $S \subseteq T$ and $T \subseteq S$. For $S \subseteq T$: “Let $x \in S$ be arbitrary. ... Therefore $x \in T$.” Split into cases wherever the definition involves “or” (a union) or “not (and)” (a complement of an intersection).

Example 2.2 (De Morgan-type identity). $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Proof. ($\subseteq$) Let $x \in A \setminus (B \cup C)$. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$. Hence $x \in A \setminus B$ and $x \in A \setminus C$.

($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$. Then $x \in A$, $x \notin B$, and $x \notin C$, so $x \notin B \cup C$. Hence $x \in A \setminus (B \cup C)$. $\square$

2.4 Induction (HW1 P2, P3)

Method 2.3. 1. **Base case:** check $P(n_0)$ directly.

2. **Write out** $P(n+1)$ by replacing every n with $n+1$. This is your target, so write it down first.

3. **Inductive step:** assume $P(n)$. Find the piece of $P(n+1)$ that is exactly $P(n)$ and substitute the hypothesis.

4. Finish with algebra.

If the *starting* index of a sum also moves with n , add and subtract the boundary terms that don't match (HW1 P3 trick).

Example 2.4 (Bernoulli's inequality, used in Week 4). For $x \geq -1$ and $n \in \mathbb{N}$: $(1+x)^n \geq 1+nx$.

Proof. Base case $n=1$: $1+x \geq 1+x$. $\checkmark$

Inductive step: assume $(1+x)^n \geq 1+nx$. Since $1+x \geq 0$, we may multiply both sides by $1+x$ without flipping the inequality:

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

because $nx^2 \geq 0$. $\square$

Note: the hypothesis $x \geq -1$ is used exactly once, to multiply an inequality by $1+x \geq 0$. Look for where each hypothesis is used.

Example 2.5 (Product identity, HW1 P2 style). For $n \geq 2$: $\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$.

Key step: $\frac{n+1}{2n} \cdot \frac{(n+1)^2-1}{(n+1)^2} = \frac{n+1}{2n} \cdot \frac{n(n+2)}{(n+1)^2} = \frac{n+2}{2(n+1)}$, using $(n+1)^2 - 1 = n(n+2)$ (difference of squares).

2.5 Contradiction and irrationality (HW1 P4, Week 2)

Method 2.6. Assume the opposite. Solve for the piece you know is irrational. Closure of $\mathbb{Q}$ under $+$, $-$, $\times$, $\div$ then forces that piece to be rational, which is a contradiction.

Example 2.7 ($\sqrt{2} \notin \mathbb{Q}$, from lecture: know this proof cold). Suppose $\sqrt{2} = p/q$ in lowest terms. Then $p^2 = 2q^2$, so p^2 is even. If $p = 2k+1$ were odd, then $p^2 = 2(2k^2+2k)+1$ would be odd. So p is even, $p = 2k$. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ and q is even too. This contradicts lowest terms.

Example 2.8 (rational $\pm$ irrational). Let $r \in \mathbb{Q}$ and $y \notin \mathbb{Q}$. Then

- (a) $r + y \notin \mathbb{Q}$ and $r - y \notin \mathbb{Q}$;
 (b) if also $r \neq 0$, then $ry \notin \mathbb{Q}$ and $y/r \notin \mathbb{Q}$.

Tool: closure of $\mathbb{Q}$. If $a = \frac{p}{q}$ and $b = \frac{s}{t}$ with $p, q, s, t \in \mathbb{Z}$ and $q, t \neq 0$, then

$$a + b = \frac{pt + sq}{qt}, \quad a - b = \frac{pt - sq}{qt}, \quad ab = \frac{ps}{qt}, \quad \frac{a}{b} = \frac{pt}{qs} \quad (\text{if } b \neq 0, \text{ i.e. } s \neq 0).$$

Each is an integer over a nonzero integer, so it is rational. This is the only fact about $\mathbb{Q}$ the proof uses.

Proof of (a), the sum. Step 1 (assume the opposite). Suppose, for contradiction, that $r + y \in \mathbb{Q}$. Give it a name: $z := r + y$, so $z \in \mathbb{Q}$.

Step 2 (solve for the irrational piece). Subtract r from both sides: $y = z - r$.

Step 3 (apply closure). Both z and r are rational, so $z - r$ is rational by closure under subtraction. Hence $y \in \mathbb{Q}$.

Step 4 (contradiction). This contradicts the hypothesis $y \notin \mathbb{Q}$. So the assumption in Step 1 was false, and $r + y \notin \mathbb{Q}$. $\square$

Proof of (a), the difference. Suppose $r - y = z \in \mathbb{Q}$. Solving for y gives $y = r - z$, which is rational by closure. This contradicts $y \notin \mathbb{Q}$, so $r - y \notin \mathbb{Q}$. $\square$

Proof of (b), the product. Step 1. Suppose, for contradiction, that $ry \in \mathbb{Q}$. Call it $w := ry$, so $w \in \mathbb{Q}$.

Step 2. Solve for y . Because $r \neq 0$ we may divide by r : $y = \frac{w}{r}$.

Step 3. w and r are rational and $r \neq 0$, so w/r is rational by closure under division. Hence $y \in \mathbb{Q}$.

Step 4. This contradicts $y \notin \mathbb{Q}$, so $ry \notin \mathbb{Q}$.

For the quotient, note $1/r \in \mathbb{Q}$ and $1/r \neq 0$, and $y/r = (1/r) \cdot y$. The product case with $1/r$ in place of r gives $y/r \notin \mathbb{Q}$. $\square$

Where is $r \neq 0$ used? Only in Step 2 of (b), to divide by r . It is really needed: if $r = 0$, then $ry = 0 \in \mathbb{Q}$, so the claim is false.

Concrete check. $3 + \sqrt{2} \notin \mathbb{Q}$: if $3 + \sqrt{2} = z \in \mathbb{Q}$, then $\sqrt{2} = z - 3 \in \mathbb{Q}$, which contradicts the previous example. Likewise $5\sqrt{2} \notin \mathbb{Q}$, because otherwise $\sqrt{2} = \frac{5\sqrt{2}}{5} \in \mathbb{Q}$.

Counterexample question: is irrational + irrational always irrational? No: $\sqrt{2} + (-\sqrt{2}) = 0$. Another one is $\sqrt{2} + (1 - \sqrt{2}) = 1$, where $1 - \sqrt{2} \notin \mathbb{Q}$ by part (a). Is irrational $\times$ irrational always irrational? No: $\sqrt{2} \cdot \sqrt{2} = 2$ and $\sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$.

Why the proof above fails here: Step 3 needs r to be rational to conclude $z - r \in \mathbb{Q}$. If r is irrational, closure says nothing, and the argument breaks down.

3 Supremum, infimum, completeness

3.1 Definitions (state these exactly)

Definition 3.1. Let $S \subseteq \mathbb{R}$ be nonempty.

- M is an **upper bound** of S if $s \leq M$ for all $s \in S$.
- $M = \sup S$ if M is an upper bound of S and no $M' < M$ is an upper bound of S . (It is the *least* upper bound.)
- $m = \inf S$ if m is a lower bound of S and no $m' > m$ is a lower bound of S .

- $\max S$ is an upper bound that *belongs to* S . If $\max S$ exists, then $\sup S = \max S$.

Theorem 3.2 (Completeness Axiom). Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. (It follows that every nonempty set bounded below has an infimum.)

Theorem 3.3 (ε -characterization of sup, the workhorse). $M = \sup S$ if and only if

- (i) $s \leq M$ for all $s \in S$, and
- (ii) for every $\varepsilon > 0$ there is $s \in S$ with $s > M - \varepsilon$.

The same holds for inf with the inequalities flipped: (ii) becomes $\forall \varepsilon > 0 \exists s \in S$ with $s < m + \varepsilon$.

Two facts used constantly:

- If t is *any* upper bound of S , then $\sup S \leq t$. (This is the “least” part.)
- $\sup S - \varepsilon$ is *not* an upper bound, so some element of S exceeds it.

3.2 Archimedean property and density

Theorem 3.4 (Archimedean property). For any $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $n > x$. Equivalently: for any $\varepsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \varepsilon$.

Proof idea (contradiction plus completeness): if $\mathbb{N}$ were bounded above, let $M = \sup \mathbb{N}$. Then $M - 1$ is not an upper bound, so some $n > M - 1$. But then $n + 1 > M$ with $n + 1 \in \mathbb{N}$, contradiction.

Theorem 3.5 (Density). If $x < y$, there is $r \in \mathbb{Q}$ with $x < r < y$, and there is an irrational z with $x < z < y$.

Proof idea for $\mathbb{Q}$: pick n with $1/n < y - x$. Let $m = \lfloor nx \rfloor + 1$. Then $nx < m \leq nx + 1 < ny$, so $x < m/n < y$.

Proof idea for irrationals (HW2 P1): either take rationals $x < r_1 < r_2 < y$ and $z = r_1 + \frac{r_2 - r_1}{\sqrt{2}}$, or apply density of $\mathbb{Q}$ to $x/\sqrt{2} < r < y/\sqrt{2}$ with $r \neq 0$ and take $z = r\sqrt{2}$.

3.3 Computing sup and inf of explicit sets

Method 3.6. 1. List the first few elements to guess the sup and inf.

- 2. If the candidate is *attained* (it is an element of S), show it is a bound. It is then the max or min, and you are done.
- 3. If it is *not attained*, show it is a bound, then use Theorem 3.3(ii) together with the Archimedean property.

Example 3.7. $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\} = \{0, \frac{1}{2}, \frac{2}{3}, \dots\}$. Then $\inf S = \min S = 0$ (at $n = 1$) and $\sup S = 1$ (not attained).

Proof that $\sup S = 1$. (i) $1 - \frac{1}{n} < 1$ for all n , so 1 is an upper bound. (ii) Let $\varepsilon > 0$. By the Archimedean property, choose n with $\frac{1}{n} < \varepsilon$. Then $1 - \frac{1}{n} > 1 - \varepsilon$, and $1 - \frac{1}{n} \in S$. By Theorem 3.3, $\sup S = 1$. $\square$

Example 3.8 (HW2 P2). $S_1 = \{\frac{1}{n}\}$: $\sup = \max = 1$, and $\inf = 0$ is not attained. $S_2 = \{\frac{(-1)^n}{n}\}$: split into odd and even n . Then $\sup S_2 = \max = \frac{1}{2}$ (at $n = 2$) and $\inf S_2 = \min = -1$ (at $n = 1$).

Example 3.9. $S = (a, b)$: $\sup S = b$ and $\inf S = a$, and neither is attained. For (ii): given $\varepsilon > 0$ (with $\varepsilon < b - a$ if needed), the point $x = b - \varepsilon/2 \in S$ satisfies $x > b - \varepsilon$.

3.4 Standard sup/inf proofs: templates

Example 3.10 ($\sup S \leq \inf T$, HW2 P3). If $s \leq t$ for all $s \in S, t \in T$: fix t . Then t is an upper bound of S , so $\sup S \leq t$. Since t was arbitrary, $\sup S$ is a lower bound of T , so $\sup S \leq \inf T$.

Example 3.11 ($\sup(A + B) = \sup A + \sup B$, HW2 P4). ($\leq$) Since $a + b \leq \sup A + \sup B$, the right side is an upper bound. ($\geq$) Given $\varepsilon > 0$, pick $a > \sup A - \frac{\varepsilon}{2}$ and $b > \sup B - \frac{\varepsilon}{2}$. Then $a + b > \sup A + \sup B - \varepsilon$. **Splitting ε into halves** is the standard trick.

Example 3.12 ($\sup(cA) = c \sup A$ for $c > 0$). Let $\alpha = \sup A$. For $a \in A$, $ca \leq c\alpha$, so $c\alpha$ is an upper bound of cA . Given $\varepsilon > 0$, pick $a \in A$ with $a > \alpha - \varepsilon/c$. Then $ca > c\alpha - \varepsilon$. (What changes if $c < 0$? Then $\sup(cA) = c \inf A$.)

Example 3.13 ($\inf S = -\sup(-S)$). $m \leq s$ for all $s \in S \iff -s \leq -m$ for all $s \in S$. So lower bounds of S are exactly the negatives of upper bounds of $-S$, and the greatest lower bound corresponds to the least upper bound.

3.5 Monotonicity of sup

If $A \subseteq B$, then $\sup A \leq \sup B$ and $\inf A \geq \inf B$. (Any upper bound of B is an upper bound of A .)

4 Absolute value

Theorem 4.1 (Properties).

1. $|xy| = |x||y|$ and $|-x| = |x|$.
2. For $a \geq 0$: $|x| \leq a \iff -a \leq x \leq a$. (**Most useful fact.**)
3. Triangle inequality: $|x + y| \leq |x| + |y|$.
4. Reverse triangle inequality: $||x| - |y|| \leq |x - y|$.
5. $|x - a| < \delta \iff a - \delta < x < a + \delta$.

Method 4.2 (Proving the triangle inequality). Add $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$ to get $-(|x| + |y|) \leq x + y \leq |x| + |y|$, then apply property 2.

Method 4.3 (Reverse triangle inequality, HW3 P1). Write $x = (x - y) + y$, so $|x| \leq |x - y| + |y|$, i.e. $|x| - |y| \leq |x - y|$. Swap x and y to get $-(|x| - |y|) \leq |x - y|$. Then property 2 gives the result.

Method 4.4 (The add-and-subtract trick). To compare a and b through c , write $a - b = (a - c) + (c - b)$ and apply the triangle inequality. This is how uniqueness of limits and the product rule are proved.

Example 4.5 (Solving equations with absolute values). Solve $|x - 3| + |x - 1| = 3$. Break the line at the points 1 and 3:

- $x < 1$: $(3 - x) + (1 - x) = 3 \Rightarrow x = \frac{1}{2}$. ✓
- $1 \leq x < 3$: $(3 - x) + (x - 1) = 2 \neq 3$. No solution.
- $x \geq 3$: $(x - 3) + (x - 1) = 3 \Rightarrow x = \frac{7}{2}$. ✓

5 Limits of sequences: the ε - N definition

Definition 5.1. $x_n \rightarrow L$ means: for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n - L| < \varepsilon$ for all $n \geq N$.

- Method 5.2** (The ε - N recipe). 1. **Guess** L (leading terms, or intuition from calculus).
 2. **Scratch work:** simplify $|x_n - L|$ to a single fraction.
 3. **Overestimate** it by something simpler, such as $\frac{C}{n}$. You make a fraction larger by making the denominator smaller or the numerator larger. You only need an upper bound.
 4. Solve $\frac{C}{n} < \varepsilon$, which gives $n > \frac{C}{\varepsilon}$, so $N > \frac{C}{\varepsilon}$ (Archimedean).
 5. **Write it forwards:** "Let $\varepsilon > 0$. Choose $N > \dots$. Then for $n \geq N$: $|x_n - L| = \dots \leq \dots < \varepsilon$."

Example 5.3 (HW3 P2). $\frac{n+2}{3n+5} \rightarrow \frac{1}{3}$.

Scratch work (not part of the proof): $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{3(3n+5)} = \frac{1}{9n+15} < \frac{1}{9n}$. We want $\frac{1}{9n} < \varepsilon$, i.e. $n > \frac{1}{9\varepsilon}$.

Proof. Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{1}{9\varepsilon}$. For $n \geq N$, $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15} < \frac{1}{9n} \leq \frac{1}{9N} < \varepsilon$. $\square$

Example 5.4 (Lecture). $\frac{2n-1}{n+1} \rightarrow 2$: $\left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1} < \frac{3}{n}$. Choose $N > \frac{3}{\varepsilon}$.

Example 5.5 (A numerator that needs bounding). $\frac{3n^2+n}{n^2+1} \rightarrow 3$: $\left| \frac{3n^2+n}{n^2+1} - 3 \right| = \frac{|n-3|}{n^2+1} \leq \frac{n+3}{n^2} \leq \frac{4n}{n^2} = \frac{4}{n}$ for $n \geq 1$ (since $3 \leq 3n$). Choose $N > \frac{4}{\varepsilon}$. **Bound the numerator above and the denominator below.**

Example 5.6 (Conjugate trick, lecture). $\frac{\sqrt{n^2+n}}{n} \rightarrow 1$: $\frac{\sqrt{n^2+n}-n}{n} = \frac{n}{n(\sqrt{n^2+n}+n)} = \frac{1}{\sqrt{n^2+n}+n} < \frac{1}{n}$. Choose $N > \frac{1}{\varepsilon}$. **Use** $\sqrt{a} - \sqrt{b} = \frac{a-b}{\sqrt{a}+\sqrt{b}}$.

Example 5.7 (Bounded over growing). $\frac{\sin n}{n} \rightarrow 0$: $\left| \frac{\sin n}{n} \right| \leq \frac{1}{n} < \varepsilon$ when $n > \frac{1}{\varepsilon}$. (You cannot use the quotient rule, because $\lim \sin n$ does not exist.)

Example 5.8 (Proving divergence). $x_n = (-1)^n$ has no limit. Suppose $x_n \rightarrow L$. Take $\varepsilon = \frac{1}{2}$ and get N . Choose an even $n \geq N$ and an odd $m \geq N$. Then $|1 - L| < \frac{1}{2}$ and $|-1 - L| < \frac{1}{2}$, so $2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < 1$, which is a contradiction.

5.1 Limits equal to $\pm\infty$

Definition 5.9. $x_n \rightarrow \infty$ means: for every M there is N such that $x_n > M$ for all $n \geq N$. Also, $x_n \rightarrow -\infty$ means $-x_n \rightarrow \infty$.

Method 5.10. Find a *simpler lower bound* $x_n \geq g(n)$ with $g(n) \rightarrow \infty$ obviously, such as $g(n) = cn$. Then choose N so that $g(n) > M$.

Example 5.11 (Lecture). $\frac{n^4-4n}{n^2} \rightarrow \infty$: for $n \geq 2$, $n^4 - 4n \geq n^3$ (since $n^3(n-1) \geq n^3 \geq 4n$ when $n \geq 2$), so $\frac{n^4-4n}{n^2} \geq n$. Given M , choose $N > \max(M, 2)$.

Example 5.12 (HW3 P3: $\infty + \text{bounded} = \infty$). Say $|y_n| \leq B$. Then $x_n + y_n \geq x_n - B$. Given K , apply $x_n \rightarrow \infty$ with the **threshold** $K + B$ to get N . For $n \geq N$, $x_n + y_n > K + B - B = K$.

6 Limit theorems (Week 3–4)

Theorem 6.1. Suppose $x_n \rightarrow x$ and $y_n \rightarrow y$. Then:

1. **Uniqueness:** limits are unique.
2. **Bounded:** a convergent sequence is bounded.
3. $ax_n + by_n \rightarrow ax + by$.
4. $x_n y_n \rightarrow xy$.
5. $x_n/y_n \rightarrow x/y$ if $y \neq 0$.
6. **Squeeze:** if $a_n \leq z_n \leq b_n$ and $a_n, b_n \rightarrow L$, then $z_n \rightarrow L$.
7. **Order:** if $x_n \leq y_n$ for all n (large), then $x \leq y$. In particular, $x_n \geq 0 \Rightarrow x \geq 0$. **Strict inequality is not preserved:** $\frac{1}{n} > 0$ but the limit is 0.

Proofs to know (short ones are quiz favorites):

- *Uniqueness:* $|x - x'| \leq |x - x_n| + |x_n - x'| < 2\varepsilon$ for every ε , so $x = x'$.
- *Bounded:* with $\varepsilon = 1$, $|x_n| < |x| + 1$ for $n \geq N$. Take $M = \max\{|x_1|, \dots, |x_{N-1}|, |x| + 1\}$.
- *Sum:* $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$, using $N = \max(N_1, N_2)$.
- *Product:* $|x_n y_n - xy| = |(x_n - x)y_n + x(y_n - y)| \leq M|x_n - x| + |x||y_n - y|$, where $|y_n| \leq M$ because convergent sequences are bounded.
- *Squeeze:* $L - \varepsilon < a_n \leq z_n \leq b_n < L + \varepsilon$.
- *Order, by contradiction:* if $x_n \geq 0$ and $x < 0$, take $\varepsilon = -x/2$. Then $x_n < x + \varepsilon = x/2 < 0$ for large n , contradiction.

6.1 The quotient rule and the “bounded away from zero” lemma

Theorem 6.2 (Lemma). If $y_n \rightarrow y \neq 0$, then there is N_0 such that $|y_n| > \frac{|y|}{2}$ for all $n \geq N_0$.

Proof. Apply the definition with $\varepsilon = \frac{|y|}{2}$. Then $|y_n| \geq |y| - |y - y_n| > |y| - \frac{|y|}{2} = \frac{|y|}{2}$ by the reverse triangle inequality. $\square$

Quotient rule sketch: by the product rule, it suffices to show $\frac{1}{y_n} \rightarrow \frac{1}{y}$. For $n \geq N_0$,

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \frac{|y - y_n|}{|y_n||y|} < \frac{2}{|y|^2} |y_n - y|,$$

which can be made $< \varepsilon$ by choosing N with $|y_n - y| < \frac{|y|^2 \varepsilon}{2}$.

6.2 More limit tools

- **Zero times bounded:** if $x_n \rightarrow 0$ and $|y_n| \leq M$, then $x_n y_n \rightarrow 0$, because $|x_n y_n| \leq M|x_n|$. (This does *not* need y_n to converge.)
- **Shifts don’t matter:** $x_n \rightarrow L \iff x_{n+1} \rightarrow L \iff x_{n+k} \rightarrow L$. Changing finitely many terms doesn’t change the limit.
- **Reciprocals and infinity:** if $x_n \rightarrow \infty$, then $\frac{1}{x_n} \rightarrow 0$. If $x_n > 0$ and $x_n \rightarrow 0$, then $\frac{1}{x_n} \rightarrow \infty$.

- **Infinity arithmetic:** $x_n \rightarrow \infty$ and $y_n \geq c > 0$ eventually $\Rightarrow x_n y_n \rightarrow \infty$. $x_n \rightarrow \infty$ and y_n bounded below $\Rightarrow x_n + y_n \rightarrow \infty$. $\infty - \infty$, $0 \cdot \infty$ and $\frac{\infty}{\infty}$ are indeterminate: for example $n - n = 0$, $(n + 1) - n = 1$, and $n^2 - n \rightarrow \infty$.
- **Comparison for infinity:** if $x_n \leq y_n$ and $x_n \rightarrow \infty$, then $y_n \rightarrow \infty$.
- **The “max of the N ’s”:** when you need two conditions at once, take $N = \max(N_1, N_2)$.
- **ε -splitting:** to get a total of $< \varepsilon$ from two pieces, make each piece $< \frac{\varepsilon}{2}$. With a constant C in front, make the piece $< \frac{\varepsilon}{C}$.

6.3 Binomial-theorem estimates (Week 4 style)

For $x \geq 0$ and $n \geq 2$: $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k \geq 1 + nx$ and also $(1+x)^n \geq \binom{n}{2} x^2 = \frac{n(n-1)}{2} x^2$. Keep only the one term you need.

Example 6.3 ($\frac{n}{2^n} \rightarrow 0$). $2^n = (1+1)^n \geq \binom{n}{2} = \frac{n(n-1)}{2}$, so for $n \geq 2$, $0 < \frac{n}{2^n} \leq \frac{2n}{n(n-1)} = \frac{2}{n-1} \rightarrow 0$. Squeeze.

Example 6.4 ($a^{1/n} \rightarrow 1$ for $a > 1$). Let $d_n = a^{1/n} - 1 > 0$. Then $a = (1+d_n)^n \geq 1 + nd_n$, so $0 < d_n \leq \frac{a-1}{n} \rightarrow 0$. For $0 < a < 1$, apply this to $\frac{1}{a}$ and take reciprocals.

Example 6.5 ($n^{1/n} \rightarrow 1$). Let $d_n = n^{1/n} - 1 \geq 0$. Then $n = (1+d_n)^n \geq \frac{n(n-1)}{2} d_n^2$, so $d_n^2 \leq \frac{2}{n-1}$ and $0 \leq d_n \leq \sqrt{\frac{2}{n-1}} \rightarrow 0$.

Example 6.6 ($\frac{a^n}{n!} \rightarrow 0$ for any a). Fix an integer $K \geq 2|a|$. For $n > K$, $\frac{|a|^n}{n!} = \frac{|a|^K}{K!} \cdot \prod_{j=K+1}^n \frac{|a|}{j} \leq \frac{|a|^K}{K!} \left(\frac{1}{2}\right)^{n-K} \rightarrow 0$.

6.4 Basic limits to memorize

1. $\frac{1}{n^p} \rightarrow 0$ for $p > 0$.
2. **Ratio of polynomials of equal degree:** the limit is the ratio of leading coefficients. (Divide the top and bottom by n^k .) If the top has lower degree, the limit is 0. If the top has higher degree, the limit is $\pm\infty$.
3. $a^n \rightarrow 0$ if $|a| < 1$. (Proof: write $\frac{1}{|a|} = 1+x$ and use Bernoulli: $|a|^n \leq \frac{1}{1+nx} \rightarrow 0$.)
4. $1 + a + \dots + a^n = \frac{1-a^{n+1}}{1-a} \rightarrow \frac{1}{1-a}$ if $|a| < 1$.
5. $\frac{\text{bounded}}{\rightarrow \infty} \rightarrow 0$, e.g. $\frac{\sin n}{n}$ and $\frac{(-1)^n}{n}$.
6. If $x_n \rightarrow L$ then $|x_n| \rightarrow |L|$ (reverse triangle inequality).

7 Monotone sequences and the Monotone Convergence Theorem

Definition 7.1. (x_n) is **increasing** if $x_n \leq x_{n+1}$ for all n , **decreasing** if $x_n \geq x_{n+1}$ for all n , and **monotone** if it is one of the two.

Method 7.2 (Showing monotonicity: three tools). 1. Show $x_{n+1} - x_n \geq 0$ (HW3 P4: common denominator, then check the sign).

2. If $x_n > 0$, show $\frac{x_{n+1}}{x_n} \geq 1$.
3. For recursive sequences, use induction: $x_n \leq x_{n+1} \Rightarrow x_{n+1} \leq x_{n+2}$.

Theorem 7.3 (Monotone Convergence Theorem). A bounded monotone sequence converges. If it is increasing and bounded above, then $x_n \rightarrow \sup\{x_n\}$.

Proof. Let $L = \sup\{x_n\}$, which exists by completeness. Let $\varepsilon > 0$. Since $L - \varepsilon$ is not an upper bound, there is N with $x_N > L - \varepsilon$. For $n \geq N$: $L - \varepsilon < x_N \leq x_n \leq L < L + \varepsilon$, so $|x_n - L| < \varepsilon$. $\square$

This proof is short and is exactly Theorem 3.3(ii) plus monotonicity. Be able to reproduce it.

Method 7.4 (Recursive sequences: the 4-step routine). 1. Compute x_1, x_2, x_3 to guess the direction and the limit.
2. Prove a bound by induction.
3. Prove monotonicity (often by induction, or using the bound).
4. By MCT, $x_n \rightarrow L$. Pass to the limit in the recursion using the limit laws, and *also* $x_{n+1} \rightarrow L$ (it is the same sequence shifted). Solve for L and use the bounds to discard wrong roots.

Example 7.5. $x_1 = 1$, $x_{n+1} = \sqrt{2 + x_n}$. Then $x_n \rightarrow 2$.

Proof. Bounded: $0 < x_n < 2$ by induction. Indeed $x_1 = 1$, and if $0 < x_n < 2$ then $0 < x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2$.

Increasing: $x_{n+1} > x_n \iff 2 + x_n > x_n^2 \iff (2 - x_n)(1 + x_n) > 0$, which holds because $0 < x_n < 2$.

Limit: by MCT, $x_n \rightarrow L$ with $1 \leq L \leq 2$. Square the recursion: $x_{n+1}^2 = 2 + x_n$. Take limits using the product and sum laws: $L^2 = 2 + L$, so $(L - 2)(L + 1) = 0$. Since $L \geq 1$, $L = 2$. $\square$

Example 7.6 (A decreasing recursive sequence). $x_1 = 3$, $x_{n+1} = \sqrt{x_n + 2}$. *Bound:* $x_n > 2$ by induction ($x_{n+1} > \sqrt{4} = 2$). *Decreasing:* $x_{n+1} < x_n \iff x_n + 2 < x_n^2 \iff (x_n - 2)(x_n + 1) > 0$, which holds. By MCT, $x_n \rightarrow L \geq 2$ with $L^2 = L + 2$, so $L = 2$.

Example 7.7 (e exists, idea only). $a_n = (1 + \frac{1}{n})^n$ is increasing and bounded above by 3 (binomial expansion compared with $\sum \frac{1}{k!} \leq 1 + \sum \frac{1}{2^{k-1}}$). So it converges, and its limit is called e . **MCT proves existence without knowing the value.**

Example 7.8 (HW3 P4, averages). If (x_n) is increasing, then so is $y_n = \frac{x_1 + \dots + x_n}{n}$. The numerator of $y_{n+1} - y_n$ is $nx_{n+1} - S_n \geq 0$, because each $x_k \leq x_{n+1}$. (Positivity is not actually needed.)

8 Preview: subsequences and Bolzano–Weierstrass (Week 5)

These may not be on Quiz 1, but you should know the statements.

- A **subsequence** is $(x_{n_k})_k$ with $n_1 < n_2 < \dots$.
- If $x_n \rightarrow L$ then every subsequence $\rightarrow L$. **So two subsequences with different limits imply divergence**, e.g. $(-1)^n$ with its even and odd terms.
- Every sequence has a monotone subsequence.
- **Bolzano–Weierstrass:** every bounded sequence has a convergent subsequence.
- **Cauchy:** $\forall \varepsilon \exists N: |x_n - x_m| < \varepsilon$ for all $n, m \geq N$. In $\mathbb{R}$, Cauchy $\iff$ convergent.

9 Common mistakes that cost points

1. **Writing ε - N proofs backwards.** Scratch work goes from the goal to N . The proof must go from “Let $\varepsilon > 0$ ” to “ $< \varepsilon$ ”.
2. **Choosing N that depends on n .** N may depend only on ε (and fixed constants).
3. **Forgetting to show that the sup is an upper bound.** Both parts (i) and (ii) are needed.
4. **Assuming the sup is in the set.** $\sup(0, 1) = 1 \notin (0, 1)$.
5. **Using limit laws when a limit doesn’t exist,** e.g. splitting $\frac{\sin n}{n}$ as $\lim \sin n \cdot \lim \frac{1}{n}$.
6. **Passing to the limit before proving convergence** in recursive sequences. MCT comes first, then solve $L = f(L)$.
7. **Multiplying an inequality by something that might be negative.**
8. **Induction without writing $P(n + 1)$ first,** and then “proving” the wrong target.
9. **Strict vs. non-strict:** limits preserve $\leq$ but not $<$.

10 Practice problems

Try these under quiz conditions. Solutions are in the next section.

Practice Problem 1. Prove by induction: $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for all $n \in \mathbb{N}$.

Practice Problem 2. Prove by induction: $2^n \geq n + 1$ for all $n \in \mathbb{N}$.

Practice Problem 3. Prove that $\sqrt{3}$ is irrational.

Practice Problem 4. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Practice Problem 5. Find sup and inf of $S = \{(-1)^n (1 - \frac{1}{n}) : n \in \mathbb{N}\}$, say whether each is attained, and prove your claim about the sup.

Practice Problem 6. Find sup and inf of $T = \{\frac{n}{n+1} : n \in \mathbb{N}\}$ with proof.

Practice Problem 7. Let A be nonempty and bounded above, and $c > 0$. Prove $\sup(cA) = c \sup A$.

Practice Problem 8. Let $A \subseteq B$ be nonempty and bounded. Prove $\inf B \leq \inf A \leq \sup A \leq \sup B$.

Practice Problem 9. Use the definition to prove $\lim_{n \rightarrow \infty} \frac{2n^2 + 1}{n^2 + 3} = 2$.

Practice Problem 10. Use the definition to prove $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

Practice Problem 11. Prove $\lim_{n \rightarrow \infty} \frac{n^2}{n+1} = \infty$ from the definition.

Practice Problem 12. Prove that if $x_n \rightarrow L$ then $|x_n| \rightarrow |L|$. Is the converse true?

Practice Problem 13. Prove that $x_n = \frac{(-1)^n n}{n+1}$ diverges.

Practice Problem 14. Prove: if $x_n \rightarrow L$ and $x_n \geq 0$ for all n , then $L \geq 0$.

Practice Problem 15. Let $x_1 = 2$ and $x_{n+1} = \frac{x_n + 3}{2}$. Prove that (x_n) converges and find its limit.

Practice Problem 16. Compute (limit laws allowed): (a) $\lim_{n \rightarrow \infty} \frac{5n^3 - n}{2n^3 + 7}$ (b) $\lim_{n \rightarrow \infty} \frac{n+1}{n^2+1}$ (c) $\lim_{n \rightarrow \infty} (\frac{2}{3})^n$
(d) $\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{2^k}$ (e) $\lim_{n \rightarrow \infty} \frac{\cos(n^2)}{\sqrt{n}}$.

Practice Problem 17 (True or false: justify or give a counterexample).

- (a) If $x_n > 0$ for all n and $x_n \rightarrow L$, then $L > 0$.
- (b) Every bounded sequence converges.
- (c) Every convergent sequence is bounded.
- (d) If $x_n y_n \rightarrow 0$ then $x_n \rightarrow 0$ or $y_n \rightarrow 0$.
- (e) If (x_n) and (y_n) both diverge, then $(x_n + y_n)$ diverges.
- (f) Every nonempty subset of $\mathbb{Q}$ that is bounded above has a sup in $\mathbb{Q}$.
- (g) If $\sup S$ exists and $\sup S \in S$, then S has a maximum.

Practice Problem 18. Let A, B be nonempty and bounded above. Prove $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

Practice Problem 19. Find sup and inf of $\{\frac{1}{n} + \frac{1}{m} : n, m \in \mathbb{N}\}$.

Practice Problem 20. Prove that if $x_n \rightarrow 0$ and (y_n) is bounded, then $x_n y_n \rightarrow 0$. Use it to find $\lim \frac{(-1)^n n}{n^2 + 1}$.

Practice Problem 21. Prove that if $x_n \rightarrow \infty$ then $\frac{1}{x_n} \rightarrow 0$ (from the definitions).

Practice Problem 22. Use the definition to prove $\lim_{n \rightarrow \infty} \frac{3n + 1}{n - 2} = 3$ (consider $n \geq 3$).

Practice Problem 23. Prove that $\frac{n}{2^n} \rightarrow 0$.

Practice Problem 24. Solve $|2x - 1| < |x + 2|$.

Practice Problem 25. Prove: if $x_n \rightarrow L > 0$, then $x_n > \frac{L}{2}$ for all sufficiently large n .

Practice Problem 26. Show that $S = \{x \in \mathbb{R} : x^2 < 5\}$ is bounded above, explain why $\sup S$ exists without knowing its value, and identify $\sup S$.

11 Solutions

1. Base case: $1 = 1^2$. Target $P(n+1)$: $1 + 3 + \cdots + (2n-1) + (2n+1) = (n+1)^2$. By the hypothesis the left side is $n^2 + 2n + 1 = (n+1)^2$. $\square$

2. Base case: $2 \geq 2$. Assume $2^n \geq n+1$. Then $2^{n+1} = 2 \cdot 2^n \geq 2(n+1) = 2n+2 \geq n+2$. $\square$

3. Suppose $\sqrt{3} = p/q$ in lowest terms, so $p^2 = 3q^2$. If $3 \nmid p$, then $p = 3k \pm 1$ and $p^2 = 9k^2 \pm 6k + 1 = 3(3k^2 \pm 2k) + 1$ is not a multiple of 3, a contradiction. So $p = 3k$, which gives $9k^2 = 3q^2$, $q^2 = 3k^2$, and so $3 \mid q$. This contradicts lowest terms. $\square$

4. ($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$, and $x \in B$ or $x \in C$. Case $x \in B$: $x \in A \cap B$. Case $x \in C$: $x \in A \cap C$. Either way x is in the union. ($\supseteq$) If $x \in A \cap B$, then $x \in A$ and $x \in B \subseteq B \cup C$. The case $x \in A \cap C$ is the same. $\square$

5. The elements are $0, \frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots$. Even n gives $1 - \frac{1}{n} \in [\frac{1}{2}, 1)$. Odd n gives $-(1 - \frac{1}{n}) \in (-1, 0]$. So $\sup S = 1$ and $\inf S = -1$, and *neither is attained*.

Proof that $\sup S = 1$: every element satisfies $(-1)^n(1 - \frac{1}{n}) \leq 1 - \frac{1}{n} < 1$, so 1 is an upper bound. Given $\varepsilon > 0$, choose an even n with $\frac{1}{n} < \varepsilon$ (Archimedean; if n is odd, use $n+1$). Then $1 - \frac{1}{n} \in S$ and $1 - \frac{1}{n} > 1 - \varepsilon$. $\square$ (The inf is symmetric, using odd n .)

6. $\frac{n}{n+1} = 1 - \frac{1}{n+1}$, which is increasing. So $\inf T = \min T = \frac{1}{2}$ (at $n = 1$), and $\sup T = 1$ is not attained. Upper bound: $\frac{n}{n+1} < 1$. Given $\varepsilon > 0$, pick n with $\frac{1}{n+1} < \varepsilon$. Then $\frac{n}{n+1} > 1 - \varepsilon$. $\square$

7. Let $\alpha = \sup A$. For $a \in A$: $a \leq \alpha$, so $ca \leq c\alpha$ (as $c > 0$), and $c\alpha$ is an upper bound of cA . Given $\varepsilon > 0$, $\frac{\varepsilon}{c} > 0$, so there is $a \in A$ with $a > \alpha - \frac{\varepsilon}{c}$. Then $ca \in cA$ and $ca > c\alpha - \varepsilon$. By Theorem 3.3, $\sup(cA) = c\alpha$. $\square$

8. Middle: pick any $a \in A$, so $\inf A \leq a \leq \sup A$. Right: $\sup B$ is an upper bound of $B \supseteq A$, so it is an upper bound of A . Hence $\sup A \leq \sup B$ ($\sup$ is the *least* upper bound). Left: the same argument with lower bounds. $\square$

9. $\left| \frac{2n^2+1}{n^2+3} - 2 \right| = \left| \frac{2n^2+1-2n^2-6}{n^2+3} \right| = \frac{5}{n^2+3} < \frac{5}{n^2} \leq \frac{5}{n}$. Let $\varepsilon > 0$ and choose $N > \frac{5}{\varepsilon}$. For $n \geq N$: $\left| \frac{2n^2+1}{n^2+3} - 2 \right| < \frac{5}{n} \leq \frac{5}{N} < \varepsilon$. $\square$

10. $\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{\sqrt{n}}$. We want $\frac{1}{\sqrt{n}} < \varepsilon \iff n > \frac{1}{\varepsilon^2}$. Let $\varepsilon > 0$ and choose $N > \frac{1}{\varepsilon^2}$. For $n \geq N$: $|\sqrt{n+1} - \sqrt{n} - 0| < \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \varepsilon$. $\square$

11. For $n \geq 1$, $n+1 \leq 2n$, so $\frac{n^2}{n+1} \geq \frac{n^2}{2n} = \frac{n}{2}$. Given M , choose $N > 2M$. For $n \geq N$: $\frac{n^2}{n+1} \geq \frac{n}{2} \geq \frac{N}{2} > M$. $\square$

12. By the reverse triangle inequality, $||x_n| - |L|| \leq |x_n - L| < \varepsilon$ for $n \geq N$, with the same N that works for $x_n \rightarrow L$. The converse is *false*: $x_n = (-1)^n$ has $|x_n| \rightarrow 1$ but diverges. (It is true when $L = 0$, since $|x_n - 0| = ||x_n| - 0|$.) $\square$

13. Suppose $x_n \rightarrow L$. The even terms are $\frac{n}{n+1} \rightarrow 1$ and the odd terms are $-\frac{n}{n+1} \rightarrow -1$. Directly: take $\varepsilon = \frac{1}{2}$. For $n \geq N$ large, even n gives $x_n > \frac{1}{2}$ and odd n gives $x_n < -\frac{1}{2}$, so these terms are more than 1 apart. But $|x_n - x_m| \leq |x_n - L| + |L - x_m| < 1$, a contradiction. (Or: two subsequences with different limits.) $\square$

14. Suppose $L < 0$. Take $\varepsilon = -\frac{L}{2} > 0$. For large n , $x_n < L + \varepsilon = \frac{L}{2} < 0$, which contradicts $x_n \geq 0$. $\square$

15. The terms are $x_1 = 2$, $x_2 = 2.5$, $x_3 = 2.75$, so the guess is increasing with limit 3. *Bound* $x_n < 3$: true for $n = 1$; if $x_n < 3$ then $x_{n+1} < \frac{3+x_n}{2} = 3$. *Increasing*: $x_{n+1} - x_n = \frac{3-x_n}{2} > 0$ by the bound. By MCT, $x_n \rightarrow L$. Taking limits in $x_{n+1} = \frac{x_n+3}{2}$ gives $L = \frac{L+3}{2}$, so $L = 3$. $\square$

16. (a) $\frac{5}{2}$ (b) 0 (lower degree on top) (c) 0 since $|\frac{2}{3}| < 1$ (d) $\frac{1}{1-1/2} = 2$ (e) 0, since $\left| \frac{\cos(n^2)}{\sqrt{n}} \right| \leq \frac{1}{\sqrt{n}} \rightarrow 0$ (squeeze).

17.

- (a) False: $\frac{1}{n} > 0$ but $\frac{1}{n} \rightarrow 0$. (You only get $L \geq 0$.)
- (b) False: $(-1)^n$. (It *does* have a convergent subsequence, by B-W.)
- (c) True: the “convergent $\Rightarrow$ bounded” theorem.
- (d) False: $x_n = 1, 0, 1, 0, \dots$ and $y_n = 0, 1, 0, 1, \dots$ have $x_n y_n = 0$, but neither converges to 0.
- (e) False: $x_n = (-1)^n$ and $y_n = -(-1)^n$ sum to 0.
- (f) False: $\{q \in \mathbb{Q} : q^2 < 2\}$ has no sup in $\mathbb{Q}$, since the sup would be $\sqrt{2}$. This is exactly why $\mathbb{R}$ needs the completeness axiom.
- (g) True: sup S is an upper bound in S , which is the definition of a maximum.

18. Let $\alpha = \sup A$, $\beta = \sup B$, $M = \max\{\alpha, \beta\}$. *Upper bound*: if $x \in A \cup B$, then $x \leq \alpha \leq M$ or $x \leq \beta \leq M$. *Least*: say $M = \alpha$. Given $\varepsilon > 0$, there is $a \in A \subseteq A \cup B$ with $a > \alpha - \varepsilon = M - \varepsilon$. $\square$

19. $\sup = \max = 2$ (at $n = m = 1$), since $\frac{1}{n} + \frac{1}{m} \leq 1 + 1$. $\inf = 0$, not attained: 0 is a lower bound since every element is > 0 . Given $\varepsilon > 0$, pick n with $\frac{1}{n} < \frac{\varepsilon}{2}$. Then $\frac{1}{n} + \frac{1}{n} < \varepsilon$, with $m = n$. $\square$

20. Let $|y_n| \leq M$ with $M > 0$. Given $\varepsilon > 0$, choose N with $|x_n| < \frac{\varepsilon}{M}$ for $n \geq N$. Then $|x_n y_n| \leq M|x_n| < \varepsilon$. Application: $\frac{(-1)^n n}{n^2+1} = (-1)^n \cdot \frac{n}{n^2+1}$, where $(-1)^n$ is bounded and $0 < \frac{n}{n^2+1} < \frac{1}{n} \rightarrow 0$. So the limit is 0. $\square$

21. Let $\varepsilon > 0$. Apply $x_n \rightarrow \infty$ with $M = \frac{1}{\varepsilon}$: there is N with $x_n > \frac{1}{\varepsilon} > 0$ for $n \geq N$. Then $0 < \frac{1}{x_n} < \varepsilon$, so $\left| \frac{1}{x_n} - 0 \right| < \varepsilon$. $\square$

22. $\frac{3n+1}{n-2} - 3 = \frac{7}{n-2} > 0$ for $n \geq 3$. For $n \geq 3$ we have $n-2 \geq \frac{n}{3}$ (equivalent to $\frac{2n}{3} \geq 2$), so $\frac{7}{n-2} \leq \frac{21}{n}$. Given $\varepsilon > 0$, choose $N > \max\{3, \frac{21}{\varepsilon}\}$. For $n \geq N$: $\left| \frac{3n+1}{n-2} - 3 \right| \leq \frac{21}{n} \leq \frac{21}{N} < \varepsilon$. $\square$

23. By the binomial theorem, $2^n \geq \binom{n}{2} = \frac{n(n-1)}{2}$ for $n \geq 2$, so $0 < \frac{n}{2^n} \leq \frac{2}{n-1}$. Given $\varepsilon > 0$, choose $N > 1 + \frac{2}{\varepsilon}$. For $n \geq N$: $\frac{n}{2^n} \leq \frac{2}{n-1} < \varepsilon$. $\square$

24. Both sides are ≥ 0 , so the inequality is equivalent to comparing squares: $(2x - 1)^2 < (x + 2)^2 \iff 3x^2 - 8x - 3 < 0 \iff (3x + 1)(x - 3) < 0 \iff -\frac{1}{3} < x < 3$. (Alternatively, split into cases at $x = \frac{1}{2}$ and $x = -2$.)

25. Take $\varepsilon = \frac{L}{2} > 0$. There is N with $|x_n - L| < \frac{L}{2}$ for $n \geq N$, so $x_n > L - \frac{L}{2} = \frac{L}{2}$. **This is the “eventually positive” lemma, used in the quotient rule.** $\square$

26. If $x \geq 3$ then $x^2 \geq 9 > 5$, so every $x \in S$ satisfies $x < 3$, and 3 is an upper bound. $S \neq \emptyset$ since $0 \in S$. By the Completeness Axiom, $\sup S$ exists, even though we haven’t computed it. In fact $\sup S = \sqrt{5}$, whose existence was proved in lecture by exactly this sup argument for $\sqrt{2}$.